

Wednesday Warm-Up: Unit 6, Momentum I and II

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1 Memory Bank

- $\vec{p} = m\vec{v}$... Definition of momentum.
- $\vec{p}_1 + \vec{p}_2 = \vec{p}_1' + \vec{p}_2'$... Momentum conservation for two objects interacting.
- $F_{\text{net}} = \Delta p / \Delta t$... Relationship between force and change in momentum.
- $F_{\text{net}} \Delta t = \Delta p$... This arrangement is known as *impulse*, the change in momentum.

2 Momentum Conservation

1. One hazard of space travel is debris left by previous missions. There are several thousand objects orbiting Earth that are large enough to be detected by radar, but there are far greater numbers of very small objects, such as flakes of paint. Calculate the force exerted by a 0.100 mg chip of paint that strikes a spacecraft window at a relative speed of 4.00×10^3 m/s, given the collision lasts 6.00×10^{-8} s.
2. A ball with an initial velocity of 10 m/s moves at an angle 60 deg above the +x-direction. The ball hits a vertical wall and bounces off so that it is moving 60 deg above the -x-direction with the same speed. In terms of m, the mass of the ball, what is the impulse delivered by the wall?
3. A 1.80 kg falcon catches a 0.650 kg dove from behind in midair. What is their velocity after impact if the falcon's velocity is initially 28.0 m/s and the dove's

velocity is 7.00 m/s in the same direction?

4. Two identical objects (such as billiard balls) have a one-dimensional collision in which one is initially motionless. After the collision, the moving object is stationary and the other moves with the same speed as the other originally had. Show that both momentum and kinetic energy are conserved.
5. A 30,000-kg freight car is coasting at 0.850 m/s with negligible friction under a hopper that dumps 110,000 kg of scrap metal into it. (a) What is the final velocity of the loaded freight car? (b) How much kinetic energy is lost?

3 Final Project Construction

1. Once you have completed the above exercises, find your lab partner and discuss what system you would like to study for your final project. (a) Sketch a basic design below, labeling all important components. (b) Make a list of tools and parts you will need. (c) What subset of tools or parts will you need from our labs?